

Protocol Change-Control Register

Protocol frozen 19 August 2026 as protocol-v1.0. The freeze provides that no item changes except by a dated, written addendum stating the reason, and that no addendum is admissible after the first test-window forecast is produced. Every addendum below was filed before that boundary was crossed: at the date of this register no test-window forecast has been generated, and Section E has not been executed beyond preflight.

Eight addenda were filed, carrying the protocol from protocol-v1.0 to protocol-v1.8. Seven concern Section D and one concern Section E evaluation design. The 17 September addendum required re-execution of an already executed selection step and changed the operative model specification. The 21 September addendum changed the frozen forecast-origin spacing and the associated HAC truncation, with consequential clarification of the HLN horizon units. The remaining amendments are execution-order, operationalization, numerical-estimation or implementation clarifications that altered no frozen threshold, candidate set or selection criterion.

Rows affected	Addendum	State	Frozen protocol (19 Aug 2026, v1.0)	Amended position after the addendum	Prompted by
SECTION D — SELECTION					
D11, D12, D13	19 Aug 2026	v1.1	Execution order of the D11 scale trigger is not stated: the protocol lists D11 after D8-D10, and "log(y+1) applied to the entire ladder" (D12) is ambiguous between re-estimating coefficients and re-running selection.	D11 evaluated on M1 residuals immediately after D5 and before D7. If triggered, D2, D3 and D5 are re-run on log(y+1); D7-D10 then execute once on the transformed scale. Order: D1, D2, D3, D4, D5, D11, (D12 to D2), D6, D7, D8, D9, D10.	Implementation review, before any Section D step was executed.
D8, D9	20 Aug 2026	v1.2	D8/D9 declare the two-stage CCF-then-AICc lag rule but do not state whether ARIMA orders and annual form are held fixed or re-selected per candidate lag.	Indicator-lag AICc selection holds the operative M1 orders and annual form fixed, as already declared for the pad at D7. Only the candidate lag varies. M2-M5 therefore differ from M1 in regressors alone, preserving the nesting relied on at E8.	Implementation review, before any Section D step was executed.
D4, D12	23 Aug 2026	v1.3	The 19 Aug addendum re-runs D2, D3 and D5 on log(y+1) but is silent on whether D4 (constant included iff d = D = 0) is re-applied on the transformed scale.	If D11 triggers D12, D4 is mechanically re-applied after the transformed-scale D2/D3 and before D5. No count-scale value of d, D or the constant decision is inherited. Sequence: D2, D3, D4, D5 on log(y+1).	Implementation-sequence review; branch not yet executed.
D5 (numerical), D7-D10	24 Aug 2026	v1.4	No iteration ceiling is declared. The transformed-scale D5 selection ran at the pmdarima default maxiter = 50 and the selected fit did not report convergence.	Downstream fixed-specification estimation from D7 onward uses method = lbfgs, maxiter = 500, applied uniformly within each decision row. Convergence status recorded for every fit; a non-converged fit at 500 pauses execution rather than being excluded or replaced. D5 artifacts retained unchanged.	Training-window convergence diagnostics only. No test-window data consulted.
D8, D9	25 Aug 2026	v1.5	"CCF identification per Box et al. (2016)" is not operationalized.	Box-Jenkins prewhitening. An auxiliary ARIMA filter is selected per indicator on the 884-day training window, then applied unchanged to the extended history and the operative response. Five candidate lags	Implementation review, before the

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				taken from the largest absolute prewhitened CCF; raw CCF is diagnostic only. Filter non-convergence pauses execution.	operative D8/D9 runs.
D9	26 Aug 2026	v1.6	C3-4 requires a week's value to enter only from its publication date, but no availability arithmetic or regressor scaling is specified.	Availability requirement lag ≥ 6 + observed publication delay, measured from week end. Stage-2 M4 fits express the search regressor in thousands (search_index / 1000), an exact reparameterisation leaving likelihood, AICc and forecasts unchanged. Candidate set, screening rule and tie rule unchanged.	Implementation review, before the operative D9 run.
D5; D7-D10 conditionally	17 Sep 2026	v1.7	The 24 Aug addendum provided that D5 "is not re-run or re-ranked", leaving the transformed-scale selection standing on a run in which 56 of 123 visited fits did not converge at maxiter = 50.	The full transformed-scale D5 stepwise search is re-executed for all seven annual-form candidates at maxiter = 500. The re-executed selection is the operative D5 result and supersedes the recorded one. Outcome: the selection changed from Fourier K = 6, ARIMA(1,1,1)(1,0,1)[7] to Fourier K = 1, ARIMA(0,1,2)(0,0,0)[7]; D7-D10 re-executed under the new operative M1.	Supervisor review of the convergence documentation. Training-window convergence status only; no test-window data consulted.
SECTION E — EVALUATION DESIGN					
E2	21 Sep 2026	v1.8	E2: origins every 7 days from 30 Nov 2025 while origin + 28 $\leq$ sample end (31 origins).	Origins every 1 day on the same start and stop rule, giving 216 origins from 30 Nov 2025 through 3 Jul 2026. Horizons unchanged at h = 7, 14, 28. Because all frozen horizons are multiples of the former 7-day spacing, every evaluated target had fallen on a single weekday; daily origins evaluate each horizon across all seven weekdays.	Deterministic calendar geometry of the frozen design. Filed before any test-window forecast.
E11	21 Sep 2026	v1.8	E11: Bartlett-kernel HAC (Newey & West, 1987), truncation $\text{ceil}(h/7) - 1$; h-step errors follow MA(h-1) (Diebold & Mariano, 1995).	Truncation changed to h - 1, i.e. 6, 13 and 27 for h = 7, 14 and 28. Direct consequence of daily origin spacing: an h-day-ahead error sequence may overlap through h-1 daily lags. Kernel and citation unchanged.	Consequence of the E2 change; same filing.
E10 (clarified)	21 Sep 2026	v1.8	E10: HLN small-sample convention applied to Clark-West; one-sided; t reference with n-1 dof. Horizon units unstated.	The HLN correction is evaluated in units of the forecast-origin sequence, so with daily origins $k = h$ rather than $\text{ceil}(h/7)$. One-sided alternative and t with n-1 dof unchanged.	Consequence of the E2 change; implementation clarification only.
E15 (clarified)	21 Sep 2026	v1.8	E15: VIFs reported for each augmented model, M2 through M5. Computation method unstated.	VIFs computed on the frozen training exogenous design with a diagnostic-only constant added, so the reported values are conventional centered VIFs. Without it, statsmodels 0.14.6 returns uncentered VIFs inflated by the column means. The constant enters no forecasting model. E15 remains descriptive with no threshold.	Implementation clarification; descriptive reporting only.

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E7, E9 (M0 clarified)	21 Sep 2026	v1.8	M0 is the same-calendar-date-previous-year benchmark (E7 MASE denominator; E9 comparator). No rule is given for 29 February.	For a 29 February target, M0 is the arithmetic mean of the observed counts on 28 February and 1 March of the preceding year. Applies equally to the E7 MASE denominator. Does not affect the present evaluation: neither the MASE interval nor the test-window targets contain 29 February.	Completeness of the benchmark definition; implementation clarification only.

Material change to a selection outcome

Only the addendum of 17 September 2026 changed a selected specification. Re-executing the transformed-scale D5 search under a uniform 500-iteration ceiling changed the operative M1 from Fourier K = 6, ARIMA(1,1,1)(1,0,1)[7] to Fourier K = 1, ARIMA(0,1,2)(0,0,0)[7]. The previously selected specification was visited in the re-execution and excluded by the search's own near-non-invertible-roots check, its seasonal structure reaching the unit circle at convergence. All seven candidates were re-estimated so that the AICc ranking was rebuilt on a common numerical setting, and D7 through D10 were re-executed under the new operative M1. No frozen candidate set, criterion, gate or tie-break was altered.

Operative specification at this register

M1: Fourier K = 1 at period 365.25, ARIMA(0,1,2)(0,0,0)[7], d = 1, D = 0, no constant, on $\log(y + 1)$.

M5: M1 with holiday regressors at pad (b, f) = (3, 2), FX at lag 84 days, and the search index at lag 42 days, scaled in thousands; nothing re-tuned.

Evaluation: expanding window, 216 daily origins from 30 November 2025 through 3 July 2026, horizons h = 7, 14, 28, coefficients re-estimated at each origin and nothing re-selected.

Declaration

The register above is a complete and accurate record of every amendment to the frozen protocol as at the date of signature. No addendum was prompted by any test-window observation, forecast or performance result. The protocol is frozen in the state protocol-v1.8 for the remainder of the study, and no further addendum is admissible once the first test-window forecast has been produced.

Signature

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Name and Date

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